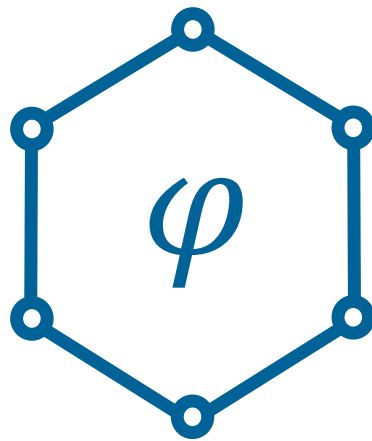


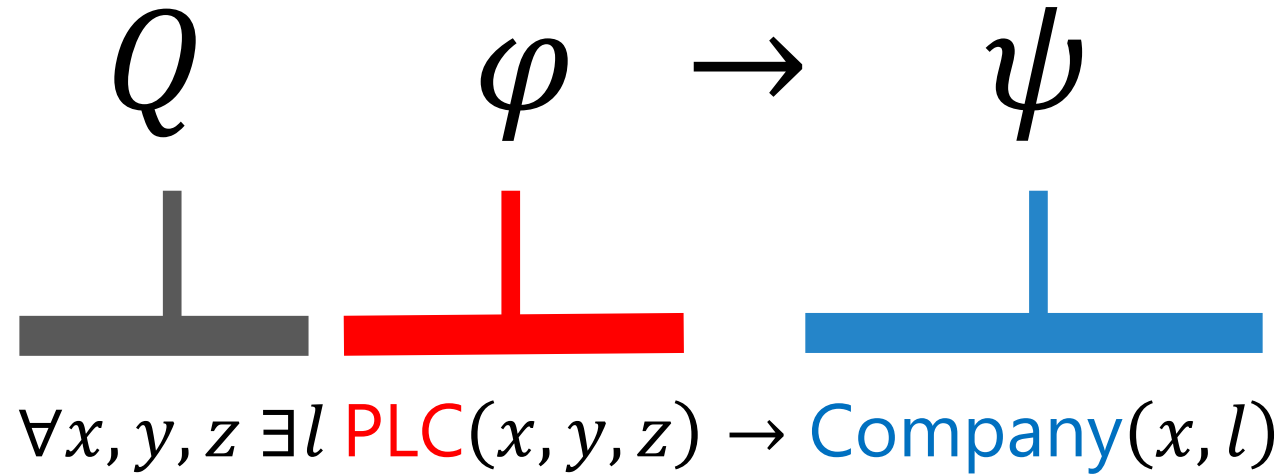


Logical Knowledge in KGs

A Unifying Framework

Primer

Emanuel Sallinger



Knowledge

"Dependency"

"Rule"

"Logical Formula"

"Ontology **Axiom**"

"Constraint"

"Named **Query**"

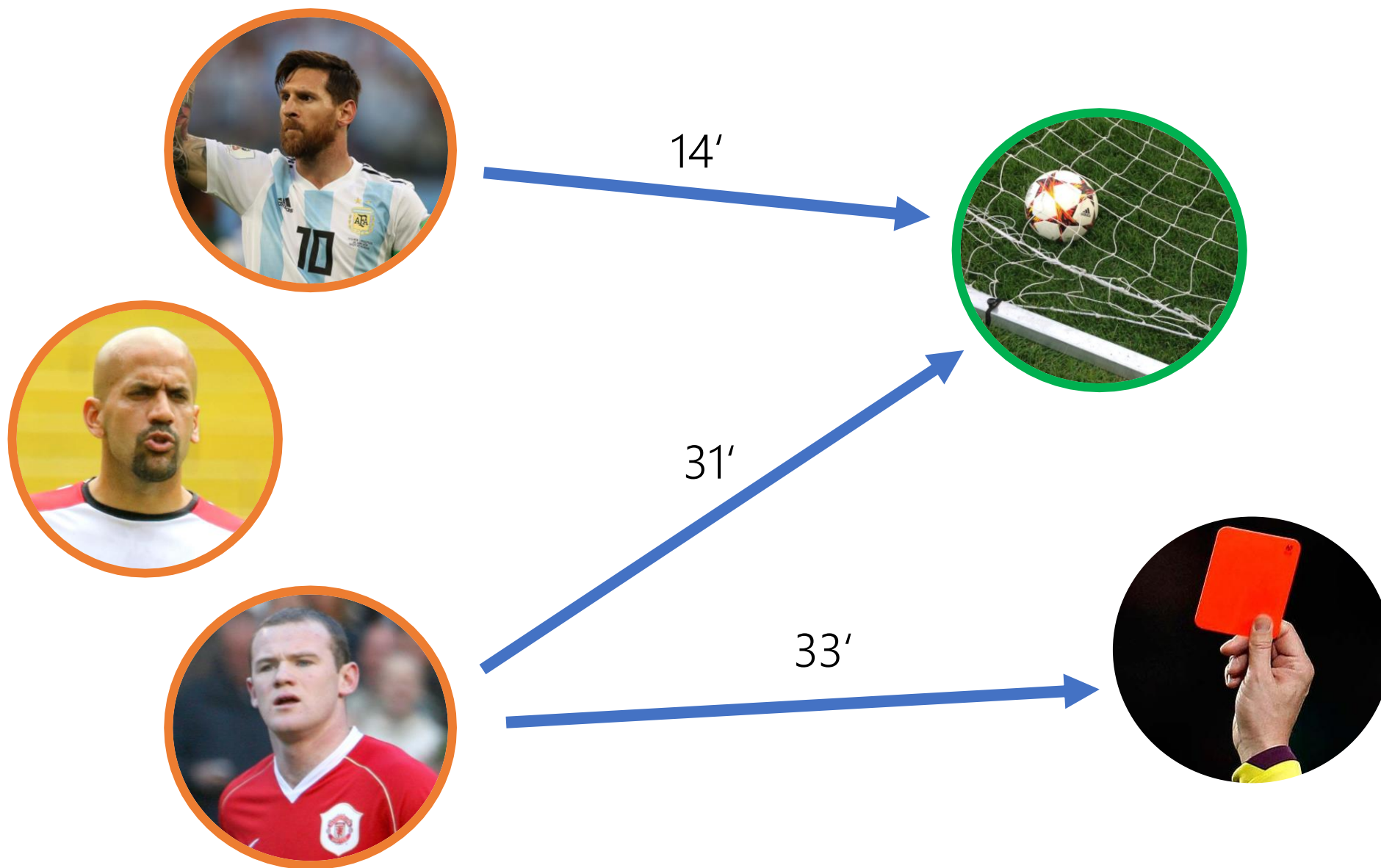


Player		
Team	No.	Name
ARG	10	Messi
ARG	3	Veron
ENG	2	Rooney



Player		
Team	No.	Name
ARG	10	Messi
ARG	3	Veron
ENG	2	Rooney

Event			
Team	No.	Type	Time
ARG	10	Goal	14'
ENG	2	Goal	31'
ENG	2	Red Card	33'



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Event			
Team	No.	Type	Time
ARG	10	Goal	14'
ENG	2	Goal	31'
ENG	2	Red Card	33'



Player		
Team	No.	Name
ARG	10	Messi
ARG	3	Veron
ENG	2	Rooney

Event			
Team	No.	Type	Time
ARG	10	Goal	14'
ENG	2	Goal	31'
ENG	2	Red Card	33'



Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'



Player		
Team	No.	Name
ARG	10	Messi
ARG	3	Veron
ENG	2	Rooney

Event			
Team	No.	Type	Time
ARG	10	Goal	14'
ENG	2	Goal	31'
ENG	2	Red Card	33'



Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'

$$\forall t, i, n \left(\text{Player}(t, i, n) \rightarrow \text{Report}(t, i, n, \text{Start}, 1) \right)$$



Player		
Team	No.	Name
ARG	10	Messi
ARG	3	Veron
ENG	2	Rooney

Event			
Team	No.	Type	Time
ARG	10	Goal	14'
ENG	2	Goal	31'
ENG	2	Red Card	33'

Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'

```
report(T,I,N,"start",1) :- player(T,I,N).  
∀t,i,n (Player(t,i,n) → Report(t,i,n,Start,1))
```



Player		
Team	No.	Name
ARG	10	Messi
ARG	3	Veron
ENG	2	Rooney

Event			
Team	No.	Type	Time
ARG	10	Goal	14'
ENG	2	Goal	31'
ENG	2	Red Card	33'



Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
ARG	10		Goal	14'
ENG	2		Goal	31'
ENG	2		Red Card	33'

$$\forall t, i, e, z \left(\text{Event}(t, i, e, z) \rightarrow \exists n \text{ Report}(t, i, n, e, z) \right)$$



Player		
Team	No.	Name
ARG	10	Messi
ARG	3	Veron
ENG	2	Rooney

Event			
Team	No.	Type	Time
ARG	10	Goal	14'
ENG	2	Goal	31'
ENG	2	Red Card	33'

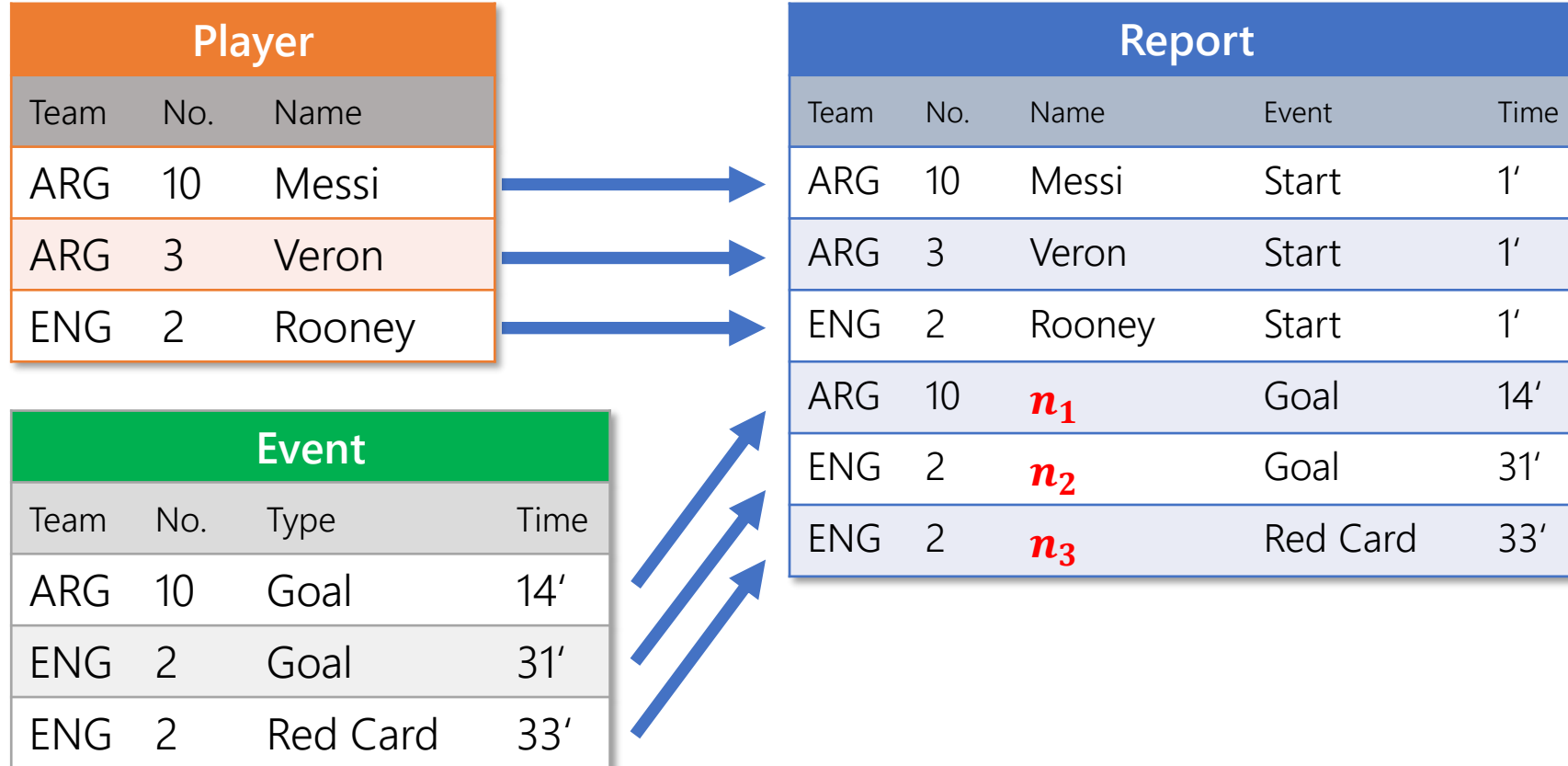


Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
ARG	10	n_1	Goal	14'
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red Card	33'

$$\forall t, i, e, z \left(\text{Event}(t, i, e, z) \rightarrow \exists n \text{ Report}(t, i, n, e, z) \right)$$



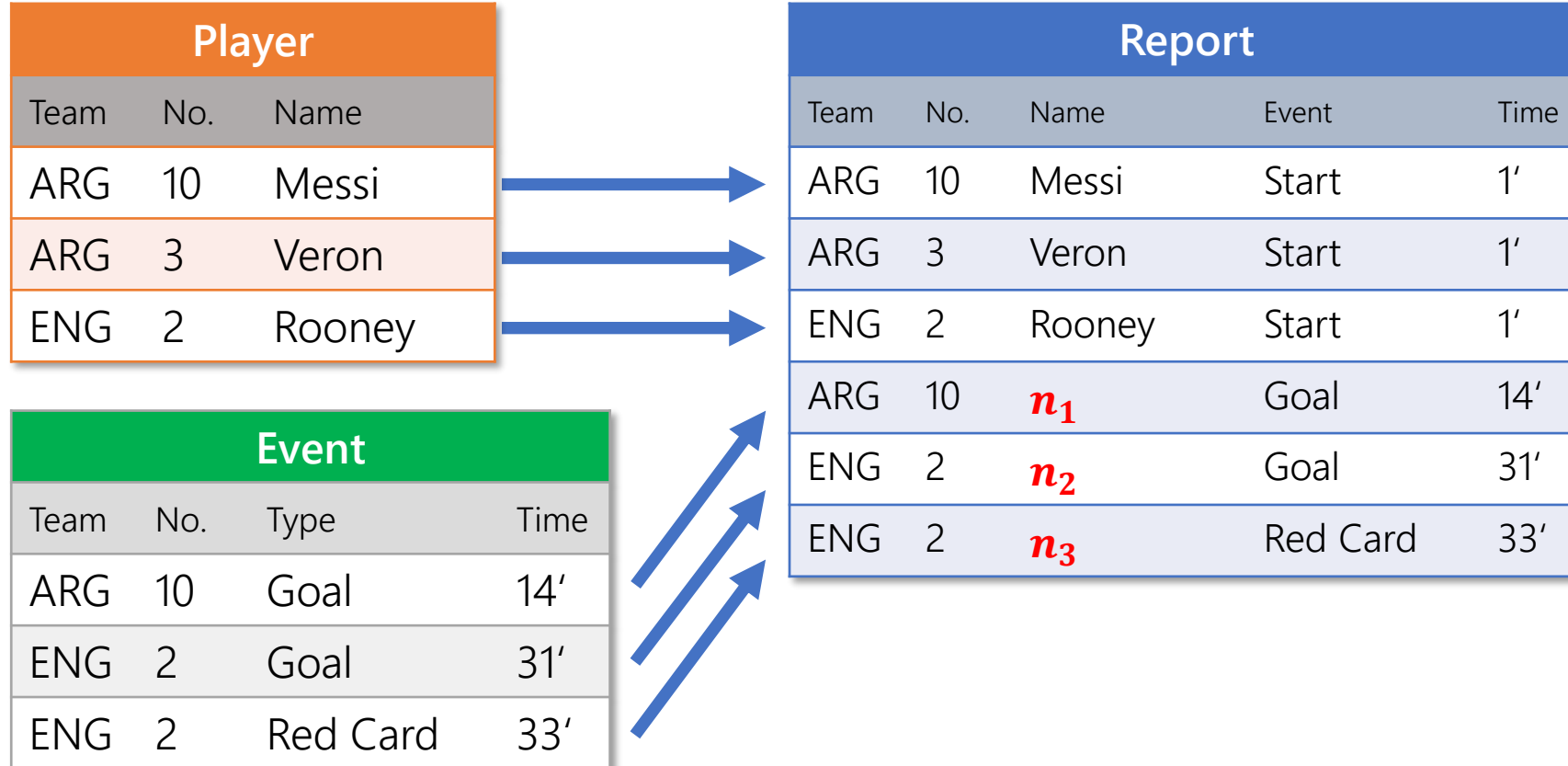
Tuple-Generating Dependency



$$\forall t, i, e, z \left(\text{Event}(t, i, e, z) \rightarrow \exists n \text{ Report}(t, i, n, e, z) \right)$$



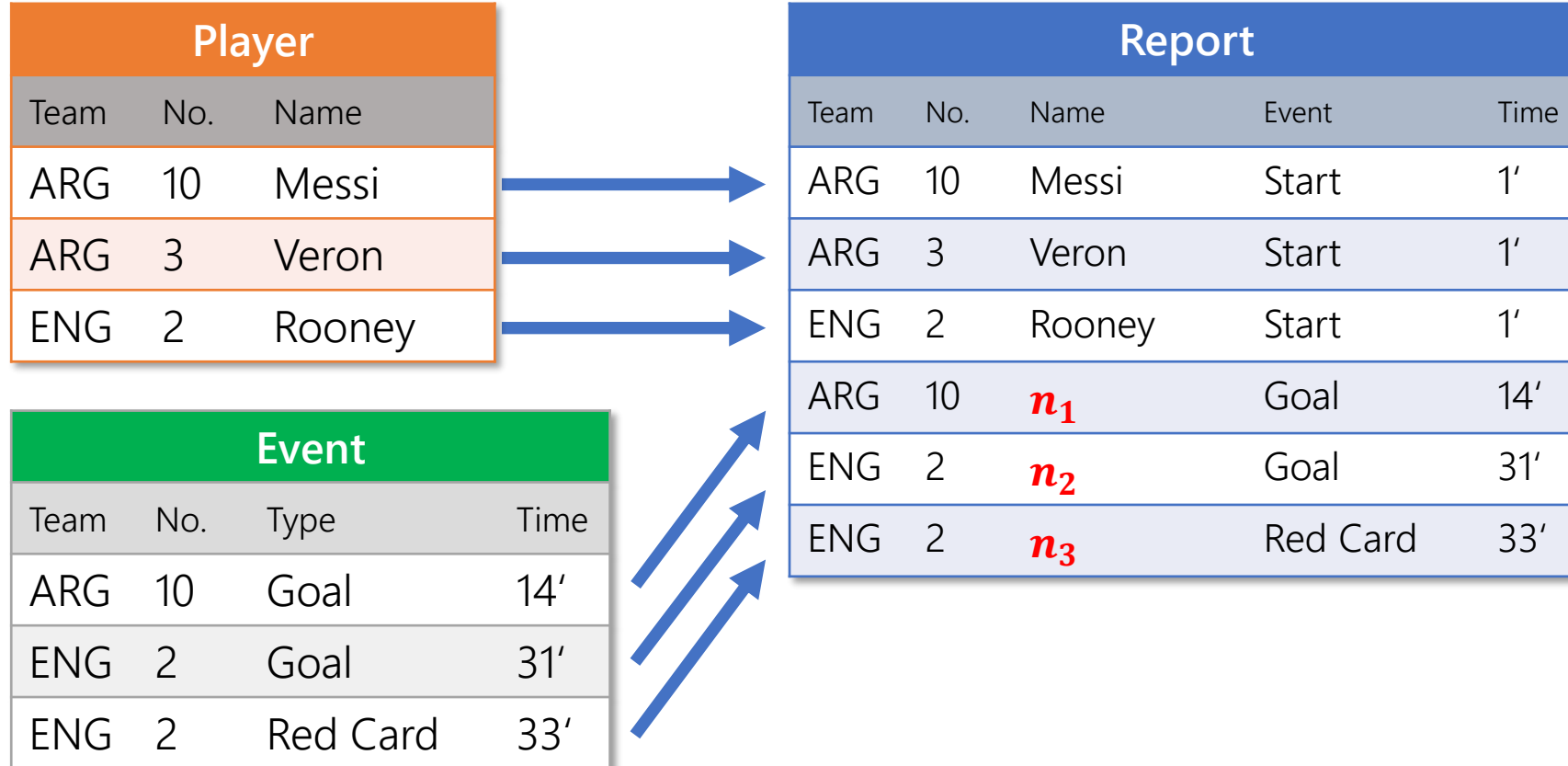
Tuple-Generating Dependency



$$\forall x (\varphi(\vec{x}) \rightarrow \exists \vec{y} \psi(\vec{x}, \vec{y}))$$



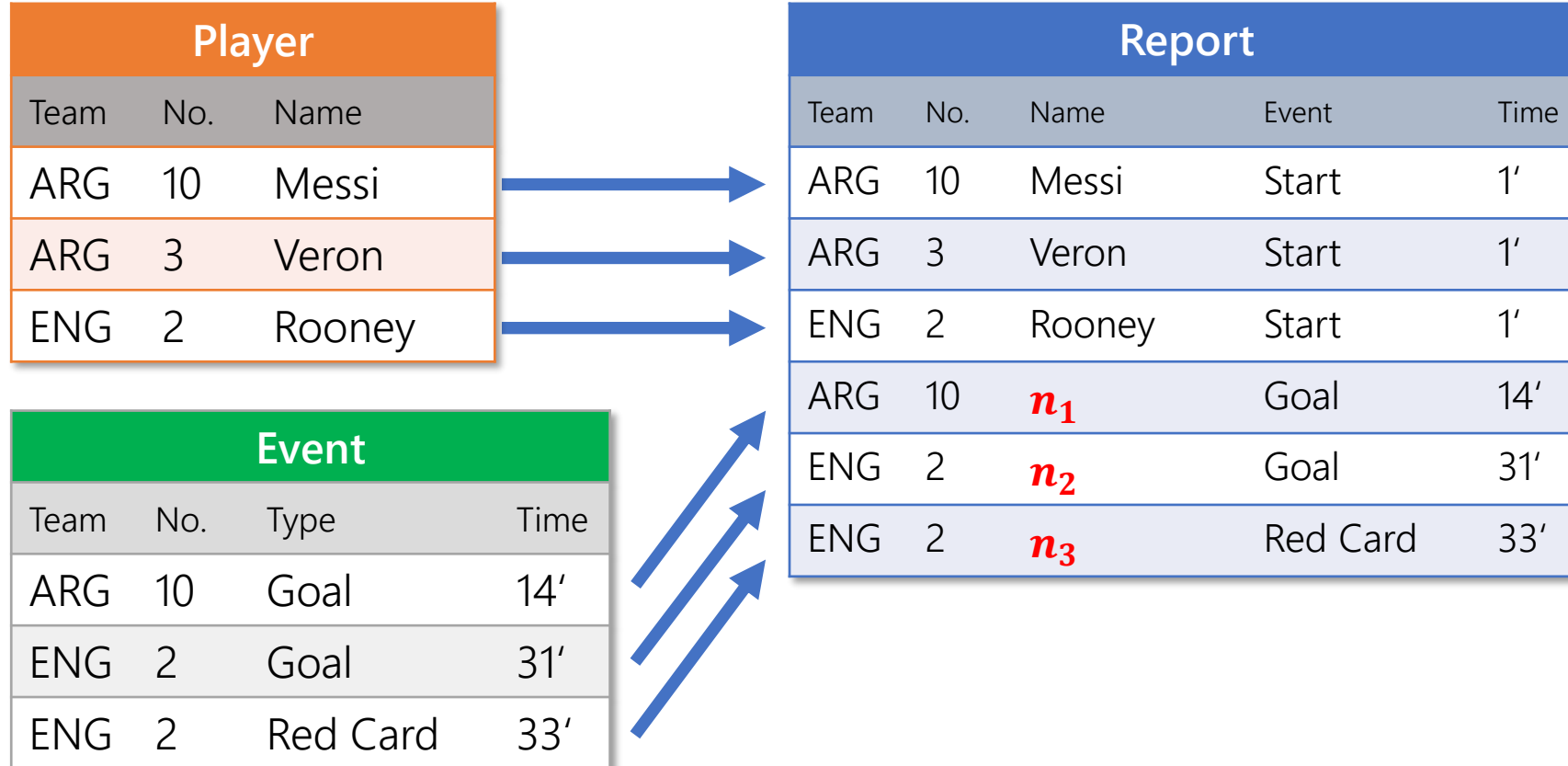
Tuple-Generating Dependency



$$\forall x (\varphi(\vec{x}) \rightarrow \exists \vec{y} \psi(\vec{x}, \vec{y}))$$



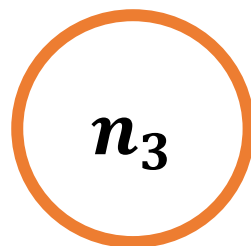
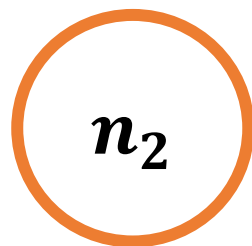
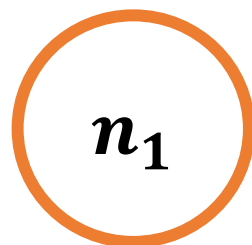
Tuple-Generating Dependency



$$\forall \vec{x} \left(\varphi(\vec{x}) \rightarrow \exists \vec{y} \, \psi(\vec{x}, \vec{y}) \right)$$



Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
ARG	10	n_1	Goal	14'
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red Card	33'



14'

31'

33'



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Report				
Team	No.	Name	Event	Time
➡ ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
➡ ARG	10	n_1	Goal	14'
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red Card	33'



Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
ARG	10	n_1	Goal	14'
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red Card	33'





$n_1 = \text{Messi}$

Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	
ARG	3	Veron	Start	
ENG	2	Rooney	Start	
ARG	10	n_1	Goal	
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red	33'



$$\forall t, i, n, e, z \ (\text{Report}(t, i, n, e, z) \wedge \text{Report}(t, i, n', e', z') \rightarrow n = n')$$



Equality-Generating Dependency

Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
ARG	10	n_1	Goal	14'
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red Card	33'



$n_1 = \text{Messi}$

$$\forall t, i, n, e, z \ (\text{Report}(t, i, n, e, z) \wedge \text{Report}(t, i, n', e', z') \rightarrow n = n')$$



Equality-Generating Dependency

Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
ARG	10	n_1	Goal	14'
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red Card	33'

$n_1 = \text{Messi}$

$$\forall x (\varphi(\vec{x}) \rightarrow x_i = x_j)$$



Equality-Generating Dependency

Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
ARG	10	n_1	Goal	14'
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red Card	33'

$n_1 = \text{Messi}$

$$\forall x \left(\varphi(\vec{x}) \rightarrow x_i = x_j \right)$$



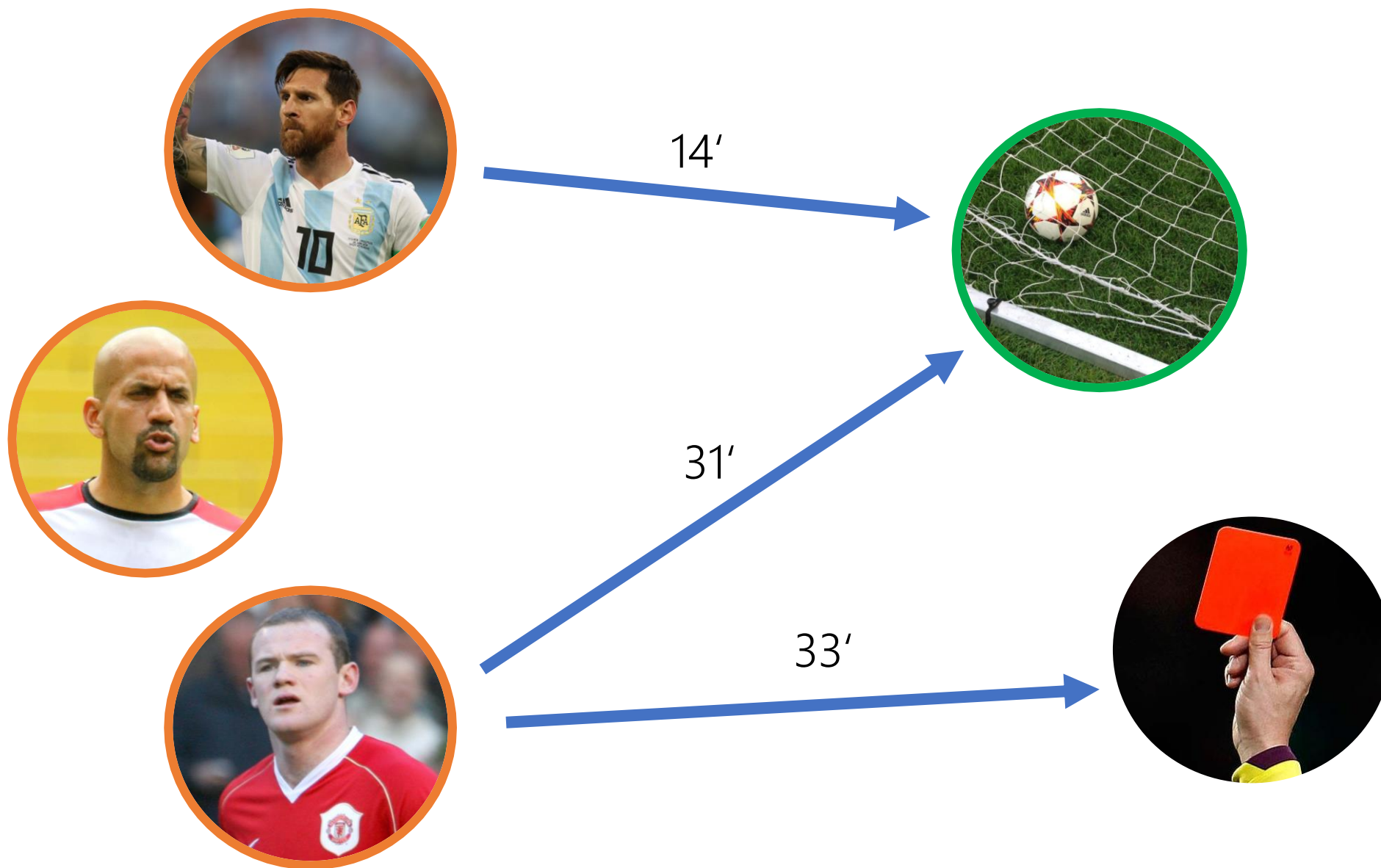
Equality-Generating Dependency

Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'
ARG	10	n_1	Goal	14'
ENG	2	n_2	Goal	31'
ENG	2	n_3	Red Card	33'



$n_1 = \text{Messi}$

$$\forall x \left(\varphi(\vec{x}) \rightarrow \mathbf{x}_i = \mathbf{x}_j \right)$$



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A Unifying Framework

Tuple-Generating Dependencies

$$\forall x \left(\varphi(\vec{x}) \rightarrow \exists \vec{y} \psi(\vec{x}, \vec{y}) \right)$$

Equality-Generating Dependencies

$$\forall x \left(\varphi(\vec{x}) \rightarrow x_i = x_j \right)$$



A Unifying Framework

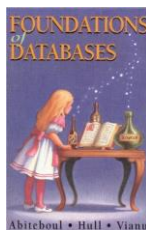
Tuple-Generating Dependencies

$$\forall \vec{x} \left(\varphi(\vec{x}) \rightarrow \exists \vec{y} \psi(\vec{x}, \vec{y}) \right)$$

Equality-Generating Dependencies

$$\forall \vec{x} \left(\varphi(\vec{x}) \rightarrow x_i = x_j \right)$$

where φ and ψ are conjunctions of atoms.



See Abitebeoul, Hull and Vianu's
"Foundations of Databases" for more details



A Unifying Framework

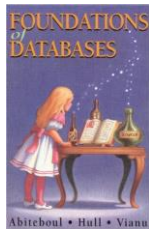
Tuple-Generating Dependencies

$$A(X,U) \text{ :- } B(X,Z), C(Z,Y).$$

Equality-Generating Dependencies

$$\forall x \left(\varphi(\vec{x}) \rightarrow x_i = x_j \right)$$

where φ and ψ are conjunctions of atoms.



See Abitebeoul, Hull and Vianu's
"Foundations of Databases" for more details



A Unifying Framework

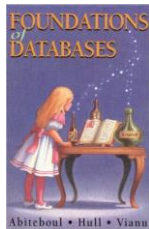
Tuple-Generating Dependencies

$$A(X,U) \text{ :- } B(X,Z), C(Z,Y).$$

Equality-Generating Dependencies

$$X=Y \text{ :- } B(X,Z), C(Z,Y).$$

where φ and ψ are conjunctions of atoms.



See Abitebeoul, Hull and Vianu's
"Foundations of Databases" for more details



Logical Knowledge in KGs:

Explicit Knowledge

$$KG = (D, K; g, l, r)$$

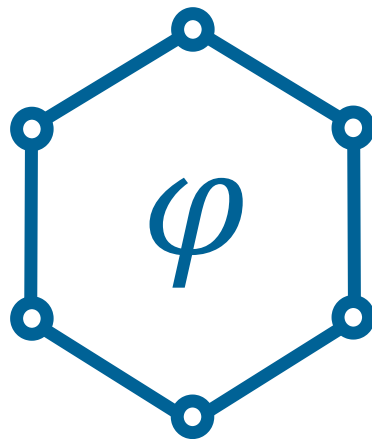
$$\mathbb{K} = \Sigma$$

$$l: \mathbb{D} \rightarrow \Sigma$$

$$r_{\Sigma}: \mathbb{D} \rightarrow \mathbb{D}$$



Query answering under **recursive tgds** is **undecidable**.



Logical Knowledge in KGs

A Unifying Framework

Primer

Emanuel Sallinger